SOFTWARE PROJECT LAB - 01

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Project Title – **Pixel++**

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Project Overview

Problem

- Digital images often suffer from:
  - Noise (unwanted distortions)
  - Blur (loss of detail)
  - Low resolution (pixelation when scaled)
- Basic processing methods may be fast but reduce image quality

Objective

- Enhance image quality using advanced algorithms
- Compare basic vs advanced techniques
- Improve performance using multithreading
- Analyze results using CSV-based benchmarking

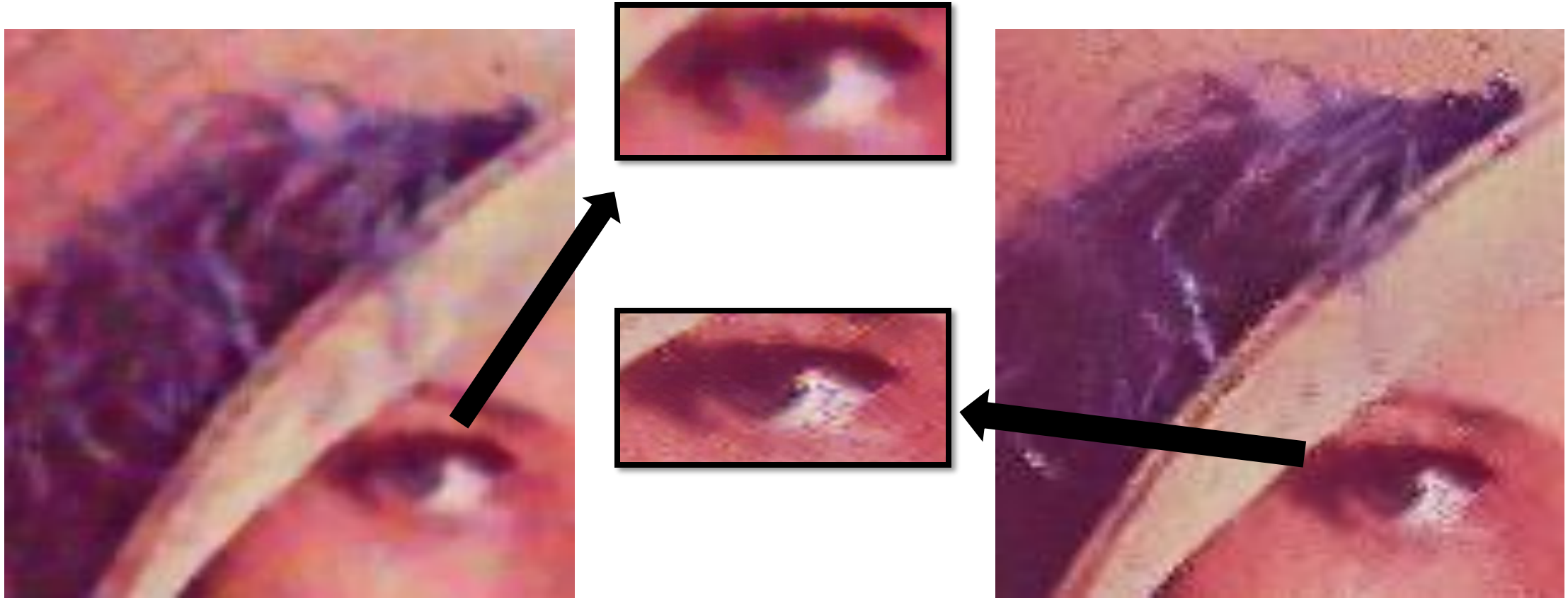
Project Evolution

Feature		During Midterm	Final
Upscaling	Bilinear Interpolation	✓	✓
	Lanczos	✗	✓
Denoising	Median Filter	✓	✓
	Non-Local Means	✗	✓
Sharpening	Laplacian Filter	✗	✓
	Unsharp Masking	✓	✓
Edge Detection	Sobel Operator	✓	✓
Performance	Multithreading	✗	✓
Analysis	CSV	✗	✓

Algorithm Overview

Feature	Basic	Advanced
Upscaling	Bilinear Interpolation	Lanczos
Denoising	Median Filter	Non-Local Means
Sharpening	Unsharp Mask	Laplacian
Edge Detection	Sobel Operator	(Enhanced quality)

Results (Denoising)



Median Filter
(Preserves less detail)

Non-Local Means
(Preserves more detail)

Results (Edge Detection)

Sobel Operator
(Static Threshold)

Sobel Operator
(Dynamic Threshold)

Optimization

Multithreading (MT)

- Divides image processing work across multiple CPU threads
- Processes different parts of the image simultaneously
- Reduces overall execution time
- Achieves significant speed improvement (up to 10x speedup)

Contiguous memory layout improves cache locality. This reduces cache misses and enhances performance during parallel processing

p11	p12	p13	p14	p15	p16	T1
p21	p22	p23	p24	p25	p26	
p31	p32	p33	p34	p35	p36	T2
p41	p42	p43	p44	p45	p46	
p51	p52	p53	p54	p55	p56	T3
p61	p62	p63	p64	p65	p66	

Performance & Benchmarking

Algorithms	Basic	MT (8 threads)	MT (16 threads)
Bilinear Interpolation	0.796872	0.132289	0.079367
Lanczos	9.550705	1.648525	1.204690
Median Filter	94.594565	13.440227	9.346646
Non-Local Means	2861.486949	658.994235	464.183417
Unsharp Masking	3.525117	0.652248	0.574929
Laplacian	2.341146	0.532913	0.424064
Sobel Operator	4.389104	0.839225	0.608131

- All algorithms are tested with **10,000 x 10,000 resolution** images

Conclusion & Challenges

Conclusion

- Successfully implemented multiple image processing features in C
- Improved image quality using advanced algorithms
- With the help of data parallelism achieved significant performance improvement
- Built a system capable of both quality enhancement and performance benchmarking

Challenges

- Efficient memory management in C during image processing
- Synchronization and coordination in multithreaded processing
- Knowing when to lock a variable and when to share it
- Balancing performance improvement with code maintainability

THANK YOU